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Efficient synthesis of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-ω-iodoalkanoates

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ABSTRACT

The efficient synthesis of four benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-ω-iodoalkanoates {benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-iodopropanoate, benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-iodobutanoate, benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-iodopentanoate, benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate} from natural or protected L-amino acids is described.

1. Introduction

ω-Iodoalkylated L-glycines equipped with protecting groups are widely used as building blocks for the preparation of numerous organic molecules. For example, naturally occurring fluorescent crosslinking amino acids, such as deoxypyridinoline **1** (Fig. 1)[1] and pentosidine **2** (Fig. 1)[2] are attractive synthetic target molecules for utilizing ω-iodoalkylated L-glycines. The collagen crosslink in **1** is formed from the adjacent lysine, and is a useful biomarker of bone resorption, and thus can be correlated with diseases such as bone cancer, osteoporosis, and arthropathies.[3] Pentosidine **2** isolated from human extracellular matrix, contains lysine and arginine side chains connected to an imidazopyridinium ring, and is found on tissue and circulating proteins from patients with renal failure.[4] The key steps in the synthesis of **1** and **2** are the palladium-catalyzed cross-coupling reaction between ω-haloalkylated L-glycine segments and haloaryl groups, and the introduction of ω-iodoalkylated L-glycine moieties onto the nitrogen atom of the pyridine ring.[5,6] Jackson et al. reported the synthesis of several types of ω-iodoalkylated L-glycines in order to study the Negishi cross-coupling with haloaryl groups for the synthesis of phenylalanine and its analogues.[7]

The convergent synthesis of protected ω-iodoalkylated L-glycines is a critical point for the synthetic study of such organic molecules. Jackson et al. synthesized some protected ω-iodoalkylated L-glycines based on the derivatization from natural L-amino acids in a couple of steps.[7] Herein, we report the efficient unified synthesis of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-ω-iodoalkanoates **3**–**6** (or ω-iodoalkylated L-glycines) from commercially available natural or protected L-amino acids (Fig. 2).

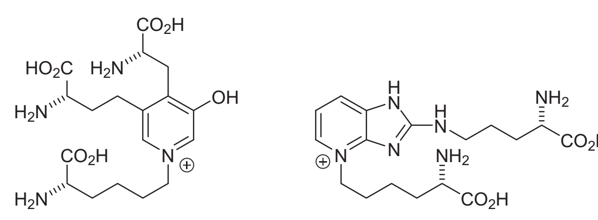

Figure 1. Structures of deoxypyridinoline **1** and pentosidine **2**.

2. Results and discussion

Our synthetic strategy took advantage of naturally derived α-amino acids (>99% ee) with the L-configuration as starting materials. In order to prevent racemization, derivatization of ω-iodoalkylated L-glycines must be conducted under mild conditions due to the acidity of the α-proton of the amino acids. The formation of iodoalkyl groups from the respective primary alcohol in amino acid systems was achieved with iodine, triphenylphosphine, and imidazole in CH_2Cl_2.[8] Thus, L-serine, 4-benzoxyl-(*S*)-3-[(*tert*-butoxycarbonyl)amino]-4-oxobutanoic acid, and 5-benzoxyl-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid were selected as the promising starting materials to be converted into the respective protected ω-iodoalkylated L-amino acids **3**–**6**.

The conversion of L-serine into benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-iodopropanoate **3** is described in Scheme 1. Protection of the carboxylic acid as the benzyl group and the amine as the *t*-butoxycarbonyl group was achieved by treatment with benzyl alcohol and di-*tert*-butyl dicarbonate ($(Boc)_2O$), respectively, to obtain alcohol **7**.[7b] The desired iodide **3** was then obtained from **7**.[8] The yield from commercially available **7** to **3** was 88% in one step, an improvement over the literature yield of 67% over two steps.[7b]

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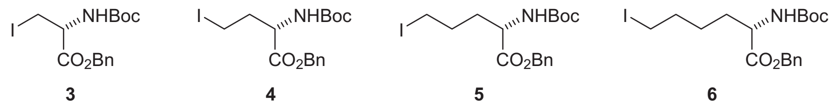
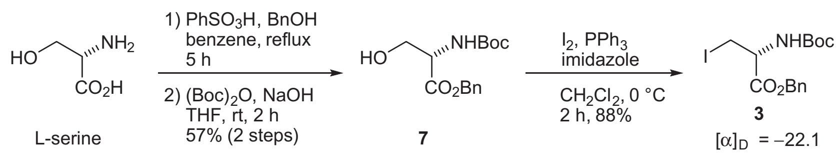
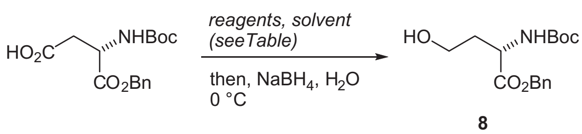


Figure 2. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-ω-iodoalkanoates **3**–**6**.



Scheme 1. Synthesis of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-iodopropanoate **3**.

Table 1
Optimization of the reduction of 4-benzoxyl-(*S*)-3-[(*tert*-butoxycarbonyl)amino]-4-oxobutanoic acid



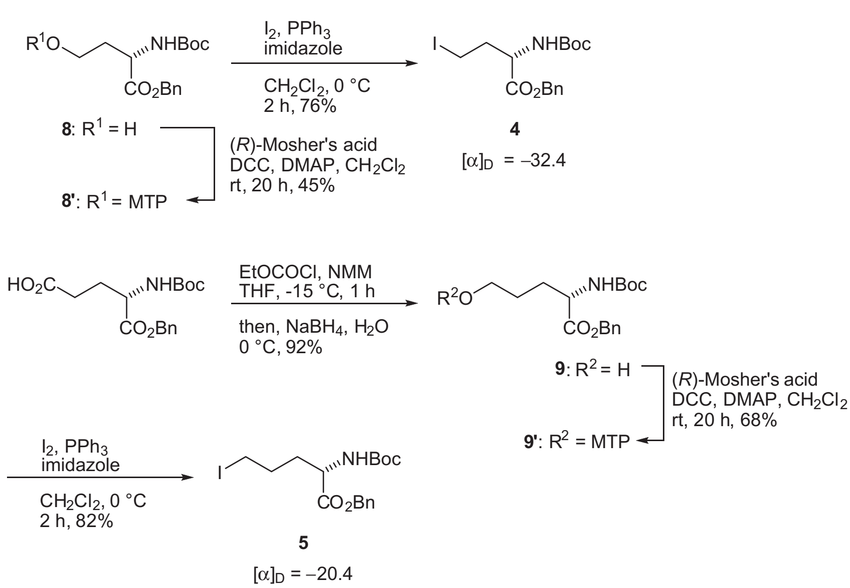
Entry	Reagents	Solvent	Temp (°C)	Time (h)	Yield (%)
1[a]	NHS, DCC	EtOAc	rt	6	64
2[b]	*i*BuOCOCl, NMM	THF	−15	1	80
3[b]	MeOCOCl, NMM	THF	−15	1	89
4[b]	EtOCOCl, NMM	THF	−15	1	94

[a] 1.1 equiv each of NHS and DCC were used. When the solution of the ester was not washed with saturated $NaHCO_3$, **8** was obtained in 58% yield. When 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC) in CH_2Cl_2 was used, the yield was 36%.

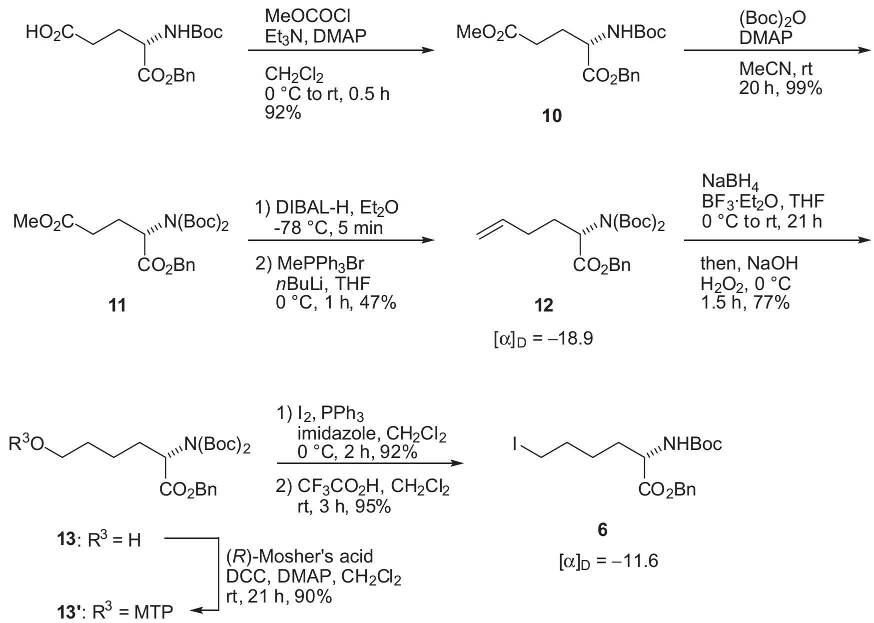
[b] 1.1 equiv each of chloroformate and NMM were used for the reaction. When triethylamine (Et_3N) was used as a base, the yield was 50%.

The key step to prepare both benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-iodobutanoate **4** and benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-iodopentanoate **5** was the chemoselective reduction of the carboxylic acid into the corresponding primary alcohol in the presence of the benzyl ester. Our approach to the reduction was to first introduce an activated ester, such as a succinimide or formic ester, onto the carboxylic acid, and then to reduce this activated ester selectively into the primary alcohol by treatment with sodium borohydride ($NaBH_4$) (Table 1).[7d] When a succinimide ester was introduced to the carboxylic acid of 4-benzoxyl-(*S*)-3-[(*tert*-butoxycarbonyl)amino]-4-oxobutanoic acid using *N*-hydroxysuccinimide (NHS) and *N*,*N*′-dicyclohexylcarbodiimide (DCC) in ethyl acetate (EtOAc), the desired benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-hydroxybutanoate **8** was obtained in 64% yield (entry 1).[7e,f] When the reduction was attempted with the chloroformate ester in the presence of *N*-methylmorpholine (NMM) as a base (entries 2–4) in tetrahydrofuran (THF),[9] the yields were improved. Ethylchloroformate was found to be the best reagent for the formation of the activated ester.

Next, the iodination of alcohol **8** was performed, which gave the desired iodide **4** in 76% yield (Scheme 2). When the hydroxyl group of **8** was converted into the corresponding α-methoxy-α-trifluoromethylphenylacetate (MTP) ester **8**′, nuclear magnetic resonance (NMR) analysis demonstrated that it was a single isomer (>99% ee). In the same manner, the optimal chemoselective reduction of 5-benzoxyl-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid gave benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)-amino]-5-hydroxypentanoate **9** in 92% yield (Scheme 2). This alcohol **9** was again converted into the corresponding MTP ester **9**′



Scheme 2. Synthesis of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-iodobutanoate **4** and benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-iodopentanoate **5**.



Scheme 3. Synthesis of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate **6**.

indicating that it was a single isomer (>99% ee). Iodination of alcohol **9** was performed and gave iodide **5** in moderate yield. According to the literature, the yields of **4** (from 4-benzoxyl-(*S*)-3-[(*tert*-butoxycarbonyl)amino]-4-oxobutanoic acid) and **5** (from 5-benzoxyl-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid) were 40% and 37%, respectively.[7d] We have thus achieved an improvement in the synthesis of **4** and **5** in 71% and 75% over two steps, respectively, utilizing efficient chemoselective reduction.

Our synthesis of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate **6** was based on extending the carbon chain via an oxidation-Wittig reaction (Scheme 3).[10] First, esterification of 5-benzoxyl-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid using methylchloroformate was carried out to give **10** in 92% yield. Adamczyk et al. suggested the direct reduction of **10** produced a complex mixture due to interference from the NH group and instability of the aldehyde.[10] A bis-Boc group was thus introduced to **10** and then **11** was reduced with diisobutylaluminium hydride (DIBAL-H) to give the aldehyde. When the Wittig reaction of this aldehyde was carried out immediately with the ylide generated from methyl triphenylphosphonium bromide and potassium *tert*-butoxide (*t*BuOK) as a base, alkene **12** was obtained in good yield. However, the specific rotation of the product was found to be $[\alpha]_D^{20} = -6.8$ (*c* 0.14, MeOH), indicating partial racemization at the α-carbon. When the ylide was prepared from methyl triphenylphosphonium bromide and *n*-butyl lithium (*n*BuLi), the specific rotation of the obtained **12** was $[\alpha]_D^{20} = -18.9$ (*c* 0.14, MeOH). Hydroboration–oxidation of **12** then gave the primary alcohol **13**. In order to investigate the racemization of the synthetic amino acid, alcohol **13** was again converted into the corresponding MTP ester **13′**. The NMR analysis suggested that no racemization (>99% ee) of **13′** occurred. The iodination followed by removal of one of the Boc groups using trifluoroacetic acid gave the desired benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate **6**. Adamczyk et al. synthesized *tert*-butyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate from 5-(*tert*-butoxyl)-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid in 13% yield in seven steps.[10] We achieved the synthesis of **6** from 5-benzoxyl-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid in seven steps with 29% overall yield.

3. Conclusion

In conclusion, the efficient synthesis of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-ω-iodoalkanoates {benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-iodopropanoate **3**, benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-iodobutanoate **4**, benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-iodopentanoate **5**, benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate **6**} with enantiomeric retention is reported starting from commercially available natural or protected L-amino acids. The overall yields for all syntheses were improved compared with literature yields. The synthesis described could be used not only for preparation of the fluorescent crosslinking amino acids **1** and **2**, but also for the synthesis of related natural products accordingly.

4. Experimental

4.1. General

Melting points were measured by an AS one ATM-01 apparatus. Optical rotations were measured on a JASCO P-2200 digital polarimeter at the sodium lamp (λ = 589 nm) D line and were recorded as follows: $[\alpha]_D^T$ (*c* g/100 mL, solvent). Infrared (IR) spectra were recorded on a JASCO FT-IR 4100 spectrometer. ^{1}H and ^{13}C NMR spectra were recorded on a JEOL Lambda 300 spectrometer (300 MHz) and a JEOL Lambda 500 spectrometer (500 MHz) in deuterated solvents. ^{1}H NMR data are reported as follows: chemical shift (δ, ppm), integration, multiplicity (s, singlet; d, doublet; t, triplet; q, quartet; m, multiplet or unresolved), coupling constants (*J*) in Hz, assignments. ^{13}C NMR data are reported in terms of chemical shift (δ, ppm). EI-MS spectra were recorded on a Shimadzu GCMS QP-5050 instrument or on a JEOL JMS-700 instrument. ESI-HRMS spectra were recorded on a JEOL JMS-T100LC instrument or on a Thermo Exactive spectrometer. FAB-MS spectra were observed on a JEOL JMS-700 instrument. Elemental analysis was performed by a Perkin Elmer PE2400 II apparatus.

All reactions were conducted under a nitrogen atmosphere with magnetic stirring using freshly distilled solvents unless otherwise indicated. Tetrahydrofuran (THF) was dried by distillation from sodium/benzophenone ketyl or stored over molecular sieves. Dichloromethane (CH_2Cl_2) and diethylether (Et_2O) were dried by distillation from calcium hydride or stored over molecular sieves. Acetonitrile (MeCN) was dried by distillation from calcium hydride. All reagents were obtained from commercial suppliers unless otherwise stated. Analytical thin layer chromatography (TLC) was performed on Silica Gel 60 F_{254} plates produced by Merck. Visualization was accomplished with UV light and phosphomolybdic acid followed by heating. Column chromatography was performed with acidic or neutral Silica Gel 60 (spherical) produced by Kanto chemicals. The enantiomeric purities (ee values) of commercially available amino acids as starting materials were >99.5%.

4.2. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-hydroxypropanoate 7

A mixture of L-serine (5.25 g, 50 mmol, 1.0 equiv), benzenesulfonic acid·hydrate (10.84 g, 62 mmol, 1.2 equiv), and benzyl alcohol (25 mL) was stirred in benzene (50 mL) with a Dean–Stark apparatus for 5 h. After removal of the remaining solvent by distillation, Et_2O (100 mL) was added to the reaction mixture with vigorous shaking for 2 h until the mixture gave a solid. Storage of the resulting mixture at 4 °C overnight yielded a solid product. After filtration, the residue was washed with cold Et_2O, and then dried. Recrystallization in hot EtOH gave L-serine benzyl ester benzenesulfonate (14.94 g, 42 mmol, >85%); ^1H NMR (300 MHz, $CDCl_3$) δ 8.12 (3H, m, NH_3^+), 7.83 (2H, d, *J* = 7.2 Hz, *o*-$PhSO_3^-$), 7.18–7.39 (8H, m, Bn, *m*,*p*-$PhSO_3^-$), 5.08 (1H, d, *J* = 12.3 Hz, Bn), 5.02 (1H, d, *J* = 12.3 Hz, Bn), 4.30 (1H, s, OH), 4.16 (1H, s, CH), 4.02 (1H, dd, *J* = 12.3, 4.5 Hz, CH_2), 3.90 (1H, dd, *J* = 12.3, 4.5 Hz, CH_2).

Next, $(Boc)_2O$ (9.16 g, 42.0 mmol, 1.3 equiv) was added to a cooled solution of L-serine benzyl ester benzenesulfonate (11.4 g, 32.3 mmol, 1.0 equiv) in THF (65 mL) and aqueous NaOH (32 mL, 1 M). The solution was stirred at room temperature for 2 h. The solution was then concentrated, and acidified with $KHSO_4$ (100 mL; 10% w/v) to pH 2–3. The aqueous phase was extracted with EtOAc (2 × 50 mL) and the combined organic layer was washed with water (3 × 33 mL). After drying over Na_2SO_4, purification on silica gel column chromatography (hexane/EtOAc = 1:0→1:1) yielded benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-hydroxypropanoate **7** as a white crystalline solid (6.42 g, 21.7 mmol, 57% (2 steps)); mp = 69–71 °C; $[\alpha]_D^{21} = -14.0$ (*c* 0.1, EtOH), $[\alpha]_D^{25} = -18.4$ (*c* 0.1, MeOH) (lit. value: $[\alpha]_D^{21} = -13.5$ (*c* 1.0, EtOH))[7b]; IR (KBr, cm^{-1}) 3442, 3373, 3091, 3066, 3034, 2960, 2927, 2856, 2179, 1949, 1748, 1717, 1500, 1456, 1367, 1344, 1250, 1164, 1051, 759, 698, 641; ^1H NMR (300 MHz, $CDCl_3$) δ 7.31–7.41 (5H, m, Bn), 5.44 (1H, br s, NH), 5.24 (1H, d, *J* = 12.0 Hz, Bn), 5.19 (1H, d, *J* = 12.0 Hz, Bn), 4.42 (1H, s, CH), 3.88–4.02 (2H, m, CH_2), 2.21 (1H, s, OH), 1.44 (9H, s, *t*Bu); ^{13}C NMR (75 Hz, $CDCl_3$) δ 170.8, 155.9, 135.3, 128.7, 128.6, 128.3, 80.5, 67.5, 63.7, 56.0, 28.4; EI-MS (*m*/*z*) calcd for $C_7H_{14}NO_3$ $[M-CO_2Bn]^+$ 160.10, found 160.15.

4.3. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-iodopropanoate 3

Triphenylphosphine (3.95 g, 15.1 mmol, 1.5 equiv) and imidazole (1.04 g, 15.3 mmol, 1.5 equiv) were dissolved in CH_2Cl_2 with stirring. Next, I_2 (3.81 g, 1.50 mmol, 1.5 equiv) was added to the solution at 0 °C. The solution was warmed to room temperature with stirring for 10 min and then cooled to 0 °C. A solution of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-hydroxypropanoate **7** (2.95 g, 10.0 mmol, 1.0 equiv) in CH_2Cl_2 (10 mL, washed with 8 mL) was also added to the solution. After stirring for 2 h, the reaction mixture was filtered through a short column of neutral silica gel eluting with hexane/Et_2O (=1:1, ca. 300 mL) and the filtrate was concentrated. Purification on silica gel column chromatography (hexane/EtOAc = 10:1→1:1) yielded benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-3-iodopropanoate **3** as a yellow solid (3.57 g, 8.8 mmol, 88%); mp = 80–81 °C; $[\alpha]_D^{20} = -18.9$ (*c* 0.1, EtOH), $[\alpha]_D^{25} = -22.1$ (*c* 0.1, MeOH) (lit. value: $[\alpha]_D^{20} = -18.6$ (*c* 1.0, EtOH))[7b]; IR (KBr, cm^{-1}) 3494, 3361, 2986, 2652, 2448, 2294, 1957, 1902, 1760, 1684, 1511, 1155, 957, 918, 860, 819, 783, 486, 453; ^1H NMR ($CDCl_3$, 300 MHz) δ 7.34–7.39 (5H, m, Bn), 5.36 (1H, br d, *J* = 7.5 Hz, NH), 5.24 (1H, d, *J* = 12.3 Hz, Bn), 5.18 (1H, d, *J* = 12.3 Hz, Bn), 4.55 (1H, m, CH), 3.61 (1H, dd, *J* = 10.2, 3.6 Hz, CH_2I), 3.55 (1H, dd, *J* = 10.2, 3.6 Hz, CH_2I), 1.45 (9 H, s, *t*Bu); ^{13}C NMR (75 Hz, $CDCl_3$) δ 169.6, 155.0, 135.0, 128.8, 128.7, 80.6, 68.1, 53.8, 28.4, 7.9; EI-MS (*m*/*z*) calcd for $C_7H_{13}INO_2$ $[M-CO_2Bn]^+$ 270.00, found 270.05; Anal. Calcd: C, 44.46; H, 4.97; N, 3.46. Found: C, 44.46; H, 5.19; N, 3.38.

4.4. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-hydroxybutanoate 8

4-Benzoxyl-(*S*)-3-[(*tert*-butoxycarbonyl)amino]-4-oxobutanoic acid (96.8 mg, 0.30 mmol, 1.0 equiv) was dissolved in THF (1.5 mL) and stirred. The solution was cooled to −15 °C, and *N*-methylmorpholine (36 μL, 0.33 mmol, 1.1 equiv) and EtOCOCl (31 μL, 0.37 mmol, 1.2 equiv) were added. The reaction mixture was then stirred for 1 h. The precipitated *N*-methylmorpholine hydrochloride was removed by filtration and washed with THF (5 mL). The filtrate was cooled to 0 °C and $NaBH_4$ (17.0 mg, 0.45 mmol, 1.5 equiv) was added, followed by the dropwise addition of water (0.3 mL). The reaction mixture was stirred at 0 °C for 0.5 h. The resulting solution was quenched with saturated aqueous NH_4Cl, stirred for 10 min, and extracted with EtOAc. The combined extracts were washed with brine, and dried over Na_2SO_4. Purification on silica gel column chromatography (hexane/EtOAc = 2:1→1:1) yielded benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-hydroxybutanoate **8** as a colorless oil (87.2 mg, 0.28 mmol, 94%); $[\alpha]_D^{27} = -40.2$ (*c* 0.1, MeOH), lit. value: $[\alpha]_D^{27} = -39.9$ (*c* 1.0, MeOH)[7e]; IR (KBr, cm^{-1}) 3355, 2935, 1684, 1511, 1369, 1261, 1155, 1053, 912, 856, 787, 745, 696, 607, 474, 444; ^1H NMR (300 MHz, $CDCl_3$) δ 7.30–7.41 (5H, m, Bn), 5.38 (1H, d, *J* = 6.6 Hz, NH), 5.19 (2H, dd, *J* = 13.4, 12.4 Hz, Bn), 4.50–4.60 (1H, m, CH), 3.59–3.74 (2H, m, CH_2O), 2.04–2.21 (1H, m, CH_2), 1.57–1.65 (1H, m, CH_2), 1.44 (9H, s, *t*Bu); ^{13}C NMR (75 MHz, $CDCl_3$) δ 172.8, 156.6, 135.3, 128.8, 128.6, 128.4, 80.6, 67.4, 58.4, 50.8, 36.2, 28.4; ESI-HRMS (*m*/*z*) calcd for $C_{16}H_{23}NO_5Na$ $[M+Na]^+$ 332.1468, found 332.1454.

4.5. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-(α-methoxy-α-trifluoromethyl-acetoxy)-butanoate 8′

At first, (*R*)-(+)-α-methoxy-α-trifluoromethylphenylacetic acid (47.5 mg, 0.20 mmol, 2.1 equiv) in CH_2Cl_2 (1.0 mL) was added to a solution of 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-hydroxybutanoate **8** (30.2 mg, 0.098 mmol, 1.0 equiv) dissolved in CH_2Cl_2 (1.3 mL) at room temperature. Next, DCC (41.2 mg, 0.20 mmol, 2.1 equiv) and DMAP (7.7 mg, 0.063 mmol, 0.6 equiv) were added to the reaction mixture. After stirring for 20 h, the mixture was filtered and the filtrate was concentrated on a rotary evaporator. Purification on silica gel column chromatography (hexane/EtOAc = 3:1) yielded 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-(α-methoxy-α-trifluoromethyl-acetoxy)-butanoate **8′** as a colorless oil (22.9 mg, 0.044 mmol, 45%); ^1H NMR (500 MHz, $CDCl_3$) δ 7.48–7.52 (2H, m, Ph), 7.30–7.41 (8H, m, Ph, Bn), 5.13 (2H, s, Bn), 5.09–5.11 (1H, m, NH), 4.38–4.44 (1H, m, CH), 4.31–4.38 (2H, m, CH_2O), 3.53 (3H, s, OMe), 2.22–2.30 (1H, m, $CHCH_2CH_2$), 2.00–2.09 (1H, m, $CHCH_2CH_2$), 1.42 (9H, s, *t*Bu); FAB-MS (*m*/*z*) $C_{21}H_{23}F_3NO_5$ $[M+2H-Boc]^+$ 426.15, found 426.21.

4.6. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-iodobutanoate 4

Triphenylphosphine (306.7 mg, 0.99 mmol, 1.7 equiv) and imidazole (73.9 mg, 1.1 mmol, 1.9 equiv) were dissolved in CH_2Cl_2 (1.8 mL) with stirring. The solution was cooled to 0 °C and I_2 (302.1 mg, 1.2 mmol, 2.1 equiv) was added. The solution was warmed to room temperature and cooled to 0 °C. A solution of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-hydroxybutanoate **8** (179.8 mg, 0.58 mmol, 1.0 equiv) in CH_2Cl_2 (0.6 mL) was also added. After stirring for 2 h, the mixture was diluted with Et_2O. The residue was filtered through a short column of neutral silica gel eluting with Et_2O. Purification on silica gel column chromatography (hexane/EtOAc = 2:1) yielded benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-4-iodobutanoate **4** as a yellow oil (184.3 mg, 0.44 mmol, 76%); $[\alpha]_D^{20} = -32.4$ (*c* 0.1, MeOH) (lit. value: -33.0 (*c* 1.0, MeOH))[11]; IR (KBr, cm^{-1}) 3356, 2979, 1756, 1681, 1518, 1445, 1370, 1298, 1251, 1153, 1052, 1028, 946, 855, 824, 789, 753, 696; ^1H NMR (300 MHz, $CDCl_3$) δ 7.31–7.41 (5H, m, Bn), 5.18 (2H, dd, *J* = 15.2, 12.2 Hz, Bn), 5.10 (1H, m, NH), 4.37 (1H, m, CH), 3.11–3.17 (2H, m, CH_2I), 2.33–2.48 (1H, m, CH_2), 2.10–2.25 (1H, m, CH_2), 1.44 (9H, s, *t*Bu); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.5, 155.4, 135.2, 128.8, 128.7, 128.4, 80.4, 67.5, 54.5, 37.1, 28.4, -0.6; ESI-HRMS (*m*/*z*) calcd for $C_{16}H_{22}NO_4INa$ $[M+Na]^+$ 442.0486, found 442.0474.

4.7. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-hydroxypentanoate 9

At first, 5-benzoxyl-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid (302.4 mg, 0.90 mmol, 1.0 equiv) was dissolved in THF (4.5 mL) and stirred. The solution was cooled to −15 °C, and *N*-methylmorpholine (109 μL, 0.99 mmol, 1.1 equiv) and EtOCOCl (82 μL, 0.99 mmol, 1.1 equiv) were added. The reaction mixture was stirred for 1 h. The precipitated *N*-methylmorpholine hydrochloride was then removed by filtration and washed with THF (10 mL). The filtrate was cooled to 0 °C and $NaBH_4$ (51.7 mg, 1.37 mmol, 1.5 equiv) was added, followed by the dropwise addition of water (0.9 mL). The reaction mixture was stirred at 0 °C for 0.5 h. The resulting solution was quenched with saturated aqueous NH_4Cl, stirred for 10 min, and extracted with EtOAc. The combined extracts were washed with brine, and dried over Na_2SO_4. Purification on silica gel column chromatography (hexane/EtOAc = 5:1) yielded benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-hydroxypentanoate **9** as a colorless oil (266.6 mg, 0.82 mmol, 92%); $[\alpha]_D^{27} = 30.7$ (*c* 0.1, MeOH), $[\alpha]_D^{25} = -1.3$ (*c* 0.1, $CHCl_3$), lit. value: $[\alpha]_D^{25} = -2.9$ (*c* 1.0, $CHCl_3$)[12]; IR (neat, cm^{-1}) 3371, 2976, 1711, 1523, 1454, 1367, 1165, 1049, 865, 752, 698; ^1H NMR (300 MHz, $CDCl_3$) δ 7.29–7.38 (5H, m, Bn), 5.24 (1H, m, NH) 5.15 (2H, dd, *J* = 19.3, 12.4 Hz, Bn), 4.36 (1H, m, CH), 3.60 (2H, t, *J* = 6.2 Hz, CH_2I), 1.54–2.16 (4H, m, CH_2CH_2), 1.42 (9H, s, *t*Bu); ^{13}C NMR (75 MHz, $CDCl_3$) δ 172.7, 155.6, 135.5, 128.7, 128.6, 128.4, 80.1, 67.2, 62.2, 60.5, 53.3, 29.5, 28.4; EI-MS (*m*/*z*) calcd for $C_9H_{18}NO_3$ $[M-CO_2Bn]^+$ 188.13, found 188.25.

4.8. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-(α-methoxy-α-trifluoromethyl-acetoxy)-pentanoate 9′

At first, (*R*)-(+)-α-methoxy-α-trifluoromethylphenylacetic acid (25.0 mg, 0.11 mmol, 1.0 equiv) in CH_2Cl_2 (1.0 mL) was added to a solution of 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-hydroxypentanoate **9** (65.6 mg, 0.20 mmol, 1.8 equiv) dissolved in CH_2Cl_2 (1.3 mL) at room temperature. Next, DCC (44.7 mg, 0.22 mmol, 2.0 equiv) and DMAP (6.7 mg, 0.055 mmol, 0.5 equiv) were added to the reaction mixture. After stirring for 20 h, the mixture was filtered and the filtrate concentrated on a rotary evaporator. Purification on silica gel column chromatography (hexane/EtOAc = 5:1) yielded 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-(α-methoxy-α-trifluoromethyl-acetoxy)-pentanoate **9′** as a colorless oil (40.2 mg, 0.075 mmol, 68%); ^1H NMR (300 MHz, $CDCl_3$) δ 7.45–7.51 (2H, m, Ph), 7.36–7.42 (3H, m, Ph), 7.28–7.36 (5H, m, Bn), 5.15 (2H, dd, *J* = 16.8, 12.0 Hz, Bn), 4.95–5.03 (1H, m, NH), 4.21–4.38 (3H, m, $CHCH_2CH_2CH_2$), 3.51 (3H, s, OMe), 1.81–1.94 (1H, m, $CHCH_2CH_2CH_2$), 1.57–1.81 (3H, m, $CHCH_2CH_2CH_2$), 1.43 (9H, s, *t*Bu); FAB-MS (*m*/*z*) calcd for $C_{22}H_{25}F_3NO_5$ $[M+2H-Boc]^+$ 440.17, found 440.24.

4.9. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-iodopentanoate 5

Triphenylphosphine (244.9 mg, 0.93 mmol, 2.0 equiv) and imidazole (63.1 mg, 0.93 mmol, 2.0 equiv) were dissolved in CH_2Cl_2 (1.35 mL) with stirring. The solution was cooled to 0 °C and I_2 (237.4 mg, 0.94 mmol, 2.0 equiv) was added. The solution was warmed to room temperature and cooled to 0 °C. A solution of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-hydroxypentanoate **9** (150.8 mg, 0.47 mmol, 1.0 equiv) in CH_2Cl_2 (0.45 mL) was also added. After stirring for 2 h, the mixture was diluted with Et_2O. The residue was filtered through a short column of neutral silica gel eluting with Et_2O. Purification on silica gel column chromatography (hexane/EtOAc = 5:1) yielded benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-5-iodopentanoate **5** as a yellow oil (165.0 mg, 0.38 mmol, 82%); $[\alpha]_D^{27} = -20.4$ (*c* 0.1, MeOH), lit. value: $[\alpha]_D^{27} = -19.9$ (*c* 1.0, MeOH)[7e]; IR (neat, cm^{-1}) 3369, 2976, 1714, 1501, 1454, 1366, 1164, 1053, 1024, 864, 751, 698, 601, 501; ^1H NMR (300 MHz, $CDCl_3$) δ 7.31–7.40 (5H, m, Bn), 5.18 (2H, dd, *J* = 20.9, 12.2 Hz, Bn), 5.04 (1H, m, NH), 4.38 (1H, m, CH), 3.08–3.21 (2H, m, CH_2I), 1.58–1.99 (4H, m, CH_2CH_2), 1.44 (9H, s, *t*Bu); ^{13}C NMR (75 MHz, $CDCl_3$) δ 172.3, 155.4, 135.3, 128.7, 128.6, 128.4, 80.1, 67.2, 52.7, 33.7, 29.2, 28.4, 21.1, 5.5; ESI-HRMS (*m*/*z*) calcd for $C_{17}H_{24}NO_4INa$ $[M+Na]^+$ 456.0642, found 456.0622.

4.10. 1-Benzyl-5-methyl-2-(*S*)-[(*tert*-butoxycarbonyl)amino]pentanedioate 10

Triethylamine (3.14 mL, 22.5 mmol, 1.5 equiv) was added to a solution of 5-benzoxyl-(*S*)-4-[(*tert*-butoxycarbonyl)amino]-5-oxopentanoic acid (5.06 g, 15.0 mmol, 1.0 equiv) dissolved in CH_2Cl_2 (68 mL). The mixture was cooled to 0 °C, after which DMAP (182.1 mg, 1.5 mmol, 0.1 equiv) followed by MeOCOCl (1.38 mL, 18.0 mmol, 1.2 equiv) were added sequentially. The solution was warmed to room temperature and stirred for 0.5 h. The reaction mixture was then diluted with CH_2Cl_2 (50 mL), washed with aqueous 1 M $NaHCO_3$ (225 mL), and extracted with CH_2Cl_2 (100 mL × 3). The organic layer was dried over Na_2SO_4. Purification on silica gel column chromatography (hexane/EtOAc = 2:1) yielded 1-benzyl-5-methyl-2-(*S*)-[(*tert*-butoxycarbonyl)amino]pentanedioate **10** as a colorless oil (4.84 g, 13.8 mmol, 92%); $[\alpha]_D^{20} = -31.3$ (*c* 0.16, MeOH); IR (KBr, cm^{-1}) 3345, 2973, 1700, 1523, 1456, 1163, 1059, 954, 903, 865, 802, 784, 758, 699, 632, 506, 435; ^1H NMR (300 MHz, $CDCl_3$) δ 7.30–7.40 (5H, m, Bn), 5.17 (2H, dd, *J* = 13.2, 12.3 Hz, Bn), 5.10 (1H, m, NH), 4.37 (1H, m, CH), 3.66 (3H, s, CO_2CH_3), 2.29–2.47 (2H, CH_2), 2.14–2.25 (1H, m, $CHCH_2CH_2$), 1.90–2.04 (1H, m, $CHCH_2CH_2$), 1.43 (9H, s, *t*Bu); ^{13}C NMR (75 MHz, $CDCl_3$) δ 173.3, 172.2, 155.5, 135.4, 128.8, 128.6, 128.4, 80.2, 67.4, 53.1, 51.9, 30.2, 28.4, 27.9; EI-MS (*m*/*z*) calcd for $C_{10}H_{18}NO_4$ $[M-CO_2Bn]^+$ 216.12, found 216.20.

4.11. 1-Benzyl-5-methyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]pentanedioate 11

At first, DMAP (336.1 mg, 2.75 mmol, 0.2 equiv) was added to a solution of 1-benzyl-5-methyl-2-(*S*)-[(*tert*-butoxycarbonyl)amino] pentanedioate **10** (4.81 g, 13.7 mmol, 1.0 equiv) dissolved in MeCN (43 mL). To this mixture, a solution of $(Boc)_2O$ (11.97 g, 54.9 mmol, 4.0 equiv) in MeCN (9 mL) was added at room temperature, and stirred for 20 h. Purification on silica gel column chromatography (hexane/EtOAc = 3:1) yielded 1-benzyl-5-methyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]pentanedioate **11** as a colorless oil (6.09 g, 13.5 mmol, 99%); $[\alpha]_D^{20} = -29.5$ (*c* 0.15, MeOH); IR (neat, cm^{-1}) 2980, 2385, 2292, 1744, 1456, 1368, 1141, 1007, 855, 782, 753, 698, 461; ^1H NMR (300 MHz, $CDCl_3$) δ 7.29–7.37 (5H, m, Bn), 5.16 (2H, dd, *J* = 12.8, 12.8 Hz, Bn), 4.97 (1H, dd, *J* = 9.6, 4.8 Hz, CH), 3.67 (3H, s, CO_2CH_3), 2.45–2.57 (1H, m, $CHCH_2CH_2$), 2.36–2.44 (2H, m, CH_2), 2.15–2.27 (1H, m, $CHCH_2CH_2$), 1.45 (18H, s, *t*Bu); ^{13}C NMR (75 MHz, $CDCl_3$) δ 173.2, 170.3, 152.1, 135.7, 128.6, 128.2, 128.1, 83.4, 66.9, 57.6, 51.8, 30.7, 28.0, 24.9; EI-MS (*m*/*z*) calcd for $C_{15}H_{26}NO_6$ $[M-CO_2Bn]^+$ 316.18, found 316.30.

4.12. Benzyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-5-hexenoate 12

Diisobutylaluminum hydride (DIBAL-H, 1 M solution in hexane, 0.84 mL, 0.84 mmol, 1.4 equiv) was added dropwise to a −78 °C cooled solution of 1-benzyl-5-methyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]pentanedioate **11** (271.9 mg, 0.60 mmol, 1.0 equiv) in Et_2O (7.2 mL) over 3 min. The reaction mixture was stirred for 5 min, then quenched with water (150 μL), and allowed to warm to room temperature. The resulting white thick solution was filtered through Celite powder and washed with Et_2O (4 mL × 3). The filtrate was concentrated and the remaining trace amounts of water were removed by azeotrope using toluene. Purification on silica gel column chromatography (hexane/EtOAc = 5:1) gave benzyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-5-oxopentanoate as a colorless oil (216.3 mg, 0.51 mmol, 85%); IR (neat, cm^{-1}) 2980, 2723, 2384, 2291, 1746, 1456, 1368, 1146, 1003, 853, 782, 753, 698, 461; ^1H NMR (300 MHz, $CDCl_3$) δ 9.77 (1H, s, CHO), 7.30–7.35 (5H, m, Bn), 5.16 (2H, dd, *J* = 14.1, 12.3 Hz, Bn), 4.92 (1H, dd, *J* = 9.4, 5.0 Hz, CH), 2.44–2.69 (3H, m, $CHCH_2CH_2$), 2.13–2.24 (1H, m, $CHCH_2CH_2$), 1.45 (18H, s, *t*Bu × 2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 201.1, 170.3, 152.2, 135.6, 128.6, 128.3, 128.1, 83.6, 67.0, 57.6, 40.6, 28.0, 22.3.

*n*BuLi (2.69 M solution in hexane, 0.19 mL, 0.52 mmol, 1.1 equiv) was added dropwise to a suspension of methyl triphenylphosphonium bromide (200.1 mg, 0.56 mmol, 1.2 equiv) in THF (6.9 mL) at 0 °C. After stirring the mixture for 0.5 h, benzyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-5-oxopentanoate (197.5 mg, 0.47 mmol, 1.0 equiv) in THF was added to the resulting orange ylide solution. The solution was stirred for 1 h at 0 °C and then quenched with saturated aqueous NH_4Cl (1.7 mL). The mixture was diluted with water (8.5 mL) and extracted with EtOAc (8.5 mL × 3). The combined organic layer was dried over Na_2SO_4. Purification on silica gel column chromatography (hexane/EtOAc = 10:1) yielded benzyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-5-hexenoate **12** as a colorless oil (107.5 mg, 0.26 mmol, 55%); $[\alpha]_D^{20} = -18.9$ (*c* 0.14, MeOH); IR (neat, cm^{-1}) 2981, 1740, 1456, 1366, 1227, 1126, 854, 752, 698; ^1H NMR (300 MHz, $CDCl_3$) δ 7.29–7.34 (5H, m, Bn), 5.74–5.85 (1H, m, CH), 5.15 (2H, dd, *J* = 15.4, 12.6 Hz, Bn), 4.89–5.08 (3H, m, $CH{=}CH_2$), 2.20–2.29 (1H, m, $CHCH_2CH_2$), 2.10–2.17 (2H, m, $CHCH_2CH_2$), 1.96–2.06 (1H, m, $CHCH_2CH_2$), 1.45 (18H, s, *t*Bu × 2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.9, 152.3, 137.5, 135.8, 128.6, 128.2, 128.1, 115.7, 83.2, 66.9, 57.8, 30.5, 29.0, 28.1; Anal. Calcd: C, 65.85; H, 7.93; N, 3.34. Found: C, 66.01; H, 8.16; N, 3.28.

4.13. Benzyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-6-hydroxyhexanoate 13

Benzyl-2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-5-hexenoate **12** (294.5 mg, 0.70 mmol, 1.0 equiv) in THF (3.4 mL) was cooled to 0 °C and $NaBH_4$ (34.7 mg, 0.92 mmol, 1.3 equiv) was added. After stirring for 10 min, $BF_3{\cdot}Et_2O$ (115 μL, 0.91 mmol, 1.3 equiv) was added to the solution. The mixture was allowed to warm to room temperature and then stirred for 21 h. The solution was then cooled to 0 °C, after which 1 M NaOH (1.05 mL, 1.05 mmol, 1.5 equiv) was added followed by 30% H_2O_2 (0.86 mL), and stirred for 1.5 h. The mixture was diluted with water (8.6 mL) and extracted with EtOAc (8.6 mL × 3). The combined organic layer was dried over Na_2SO_4. Purification on silica gel column chromatography (hexane/EtOAc = 1:1) yielded benzyl 2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-6-hydroxyhexanoate **13** as a colorless oil (237.5 mg, 0.54 mmol, 77%); $[\alpha]_D^{20} = -16.0$ (*c* 0.13, MeOH); IR (neat, cm^{-1}) 3540, 2979, 1744, 1457, 1368, 1143, 994, 854, 782, 755, 698, 602, 463; ^1H NMR (300 MHz, $CDCl_3$) δ 7.29–7.35 (5H, m, Bn), 5.15 (2H, dd, *J* = 16.8, 12.3 Hz, Bn), 4.89 (1H, dd, *J* = 9.6, 5.1 Hz, CH), 3.64 (2H, t, *J* = 6.6 Hz, CH_2O), 2.12–2.19 (1H, m, $CHCH_2CH_2CH_2CH_2OH$), 1.89–1.97 (1H, m, $CHCH_2CH_2CH_2CH_2OH$), 1.55–1.65 (2H, m, $CHCH_2CH_2CH_2CH_2OH$), 1.40–1.50 (2H, m, $CHCH_2CH_2CH_2CH_2OH$), 1.47 (18H, s, *t*Bu × 2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.9, 152.5, 135.8, 128.6, 128.2, 128.1, 83.2, 66.9, 62.7, 58.2, 32.4, 29.3, 28.1, 22.5.

4.14. Benzyl 2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-6-(α-methoxy-α-trifluoromethyl-acetoxy)-hexanoate 13′

At first, (*R*)-(+)-α-methoxy-α-trifluoromethylphenylacetic acid (23.0 mg, 0.096 mmol, 2.0 equiv) in CH_2Cl_2 (0.8 mL) was added to a solution of benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-hydroxyhexanoate **13** (21.1 mg, 0.048 mmol, 1.0 equiv) dissolved in CH_2Cl_2 (0.6 mL) at room temperature. Next, DCC (20.3 mg, 0.098 mmol, 2.0 equiv) and DMAP (4.2 mg, 0.072 mmol, 0.5 equiv) were added to the reaction mixture. After stirring the reaction mixture for 21 h, the mixture was filtered and the filtrate was concentrated on a rotary evaporator. Purification on silica gel column chromatography (hexane/EtOAc = 5:1) yielded 2-(*S*)-[(*tert*-butoxycarbonyl)-amino]-6-(α-methoxy-α-trifluoromethyl-acetoxy)-hexanoate **13’** as a colorless oil (28.3 mg, 0.043 mmol, 90%). ^1H NMR (500 MHz, $CDCl_3$) δ 7.48–7.53 (2H, m, Ph), 7.38–7.41 (3H, m, Ph), 7.28–7.36 (5H, m, Bn), 5.14 (2H, dd, *J* = 12.6, 7.5 Hz, Bn), 4.86 (1H, q, *J* = 3.0 Hz, CH), 4.31 (2H, t, *J* = 4.0 Hz, CH_2O), 3.54 (3H, s, OMe), 2.01–2.18 (1H, m, $CHCH_2$), 1.88–1.97 (1H, m, $CHCH_2$), 1.67–1.80 (2H, m, $CHCH_2CH_2CH_2$), 1.38–1.50 (2H, m, $CHCH_2CH_2CH_2$), 1.43 (18H, s, *t*Bu × 2); FAB-MS (*m*/*z*) calcd for $C_{23}H_{27}F_3NO_5$ $[M+3H-2Boc]^+$ 454.18, found 454.28.

4.15. Benzyl 2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-6-iodohexanoate

Triphenylphosphine (1.23 g, 4.7 mmol, 2.0 equiv) and imidazole (320.7 mg, 4.7 mmol, 2.0 equiv) were dissolved in CH_2Cl_2 (7.1 mL) with stirring. Next, I_2 (1.19 g, 4.7 mmol, 2.0 equiv) was added to the solution at 0 °C. The solution was warmed to room temperature and then cooled to 0 °C. A solution of benzyl 2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-6-hydroxyhexanoate **13** (1.03 g, 2.35 mmol, 1.0 equiv) in CH_2Cl_2 (4.3 mL) was then added to the solution. After stirring for 2 h, the mixture was diluted with Et_2O. The residue was filtered through a short column of neutral silica gel by eluting with Et_2O. Purification on silica gel column chromatography (hexane/EtOAc = 5:1) yielded benzyl 2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-6-iodohexanoate as a yellow oil (1.18 g, 2.2 mmol, 92%); $[\alpha]_D^{20} = -19.0$ (*c* 0.1, MeOH), $[\alpha]_D^{27} = -14.7$ (*c* 0.1, MeOH); IR (neat, cm^{-1}) 2978, 1793, 1752, 1498, 1456, 1367, 1135, 996, 854, 754, 697, 597, 464; ^1H NMR (300 MHz, $CDCl_3$) δ 7.31–7.34 (5H, m, Bn), 5.15 (2H, dd, *J* = 16.2, 12.6 Hz, Bn), 4.88 (2H, dd, *J* = 9.6, 5.1 Hz, CH), 3.18 (2H, t, *J* = 6.8 Hz, CH_2I), 2.08–2.17 (1H, m, $CH_2CH_2CH_2CH_2I$), 1.82–2.00 (3H, m, $CHCH_2$-

$CH_2CH_2CH_2I$), 1.41–1.48 (2H, m, $CHCH_2CH_2CH_2CH_2I$), 1.45 (18H, s, *t*Bu × 2); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.7, 152.3, 135.8, 128.6, 128.2, 128.1, 83.3, 66.9, 58.1, 33.1, 28.5, 28.1, 27.3, 6.5; EI-MS (*m*/*z*) calcd for $C_{15}H_{27}INO_4$ $[M-CO_2Bn]^+$ 412.10, found 412.25.

4.16. Benzyl 2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate 6

Trifluoroacetic acid (44.5 μL, 0.58 mmol, 1.9 equiv) was added to a solution of benzyl 2-(*S*)-[bis-(*tert*-butoxycarbonyl)amino]-6-iodohexanoate (161.5 mg, 0.30 mmol, 1.0 equiv) dissolved in CH_2Cl_2 (4 mL). The reaction mixture was stirred for 3 h at room temperature. The mixture was then diluted with CH_2Cl_2 (15 mL) and washed with 10% aqueous NaOH (8 mL), and extracted with CH_2Cl_2 (8 mL × 3). The organic layer was dried over Na_2SO_4. Purification on silica gel column chromatography (hexane/EtOAc = 5:1) yielded benzyl-2-(*S*)-[(*tert*-butoxycarbonyl)amino]-6-iodohexanoate **6** as a colorless oil (125.4 mg, 0.28 mmol, 95%); $[\alpha]_D^{20} = -11.6$ (*c* 0.1, MeOH), $[\alpha]_D^{27} = -8.6$ (*c* 0.1, MeOH); IR (neat, cm^{-1}) 3369, 2976, 1714, 1501, 1455, 1366, 1165, 1051, 1026, 867, 751, 698, 600, 498; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.33–7.40 (5H, m, Bn), 5.18 (2H, dd, *J* = 25.4, 12.2 Hz, Bn), 5.04 (1H, m, NH), 4.35 (1H, m, CH), 3.11 (2H, t, *J* = 6.9 Hz, CH_2I), 1.73–1.87 (3H, m, $CHCH_2CH_2CH_2CH_2I$), 1.57–1.70 (1H, m, $CHCH_2CH_2CH_2CH_2I$), 1.37–1.50 (2H, m, $CHCH_2CH_2CH_2CH_2I$), 1.44 (9H, s, *t*Bu); ^{13}C NMR (75 MHz, $CDCl_3$) δ 172.6, 155.4, 135.4, 128.7, 128.6, 128.5, 80.1, 67.2, 53.3, 32.9, 31.7, 28.4, 26.2, 6.2; ESI-HRMS (*m*/*z*) calcd for $C_{18}H_{26}O_4NINa$ $[M+Na^+]$ 470.0804, found 470.0796.

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